

Carmen Lucca DiMario

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EDUCATION

Embry-Riddle Aeronautical University (ERAU)

Master of Science in Computer Science

Bachelor of Science in Computer Science

GPA: 4.0

GPA: 3.97

Graduation:

May 2027

May 2026

TECHNICAL SKILLS

Clearance: Held interim DoD Secret (inactive) | U.S. Citizen | FAA Part 107 Remote Pilot

Programming Languages: C, C++, Python, Java, Go, TypeScript, JavaScript, SQL, GNU ASM, Bash

Embedded Systems: STM32 (ARM Cortex-M), I2C/SPI, Clang/ARMcc toolchains, PX4/XRCE-DDS, NVIDIA Jetson, ROS2, TensorRT

ML & AI: PyTorch, Stable-Baselines3, Gymnasium, NumPy

Tools and Technologies: UNIX, Linux, REST APIs, PostgreSQL, Docker, Git, GitLab, Jenkins, AWS

PROFESSIONAL EXPERIENCE

Garmin: Embedded Software Development Intern

May 2026 - August 2026

- Implemented Clang compiler support as a replacement for legacy ARMcc toolchains, modernizing the build for avionics firmware
- Extended embedded library support for 64-bit architectures, resolving data type alignment and pointer-width issues to broaden platform compatibility
- Developed a system test suite for a proprietary avionics communication interface, validating real-time data transmission and protocol correctness

JRC Integrated Systems: RISE Intern/CO-OP Participant

June 2025 - May 2026

- Upgraded Accuro's PETSc integration across a major version boundary to enable GPU-accelerated solvers for the differential equations at the core of its simulation
- Rebuilt the test suite for Accuro, a radiation-effects semiconductor simulator, from shell scripts into a parallel Python framework with selectable test subsets, cutting a multi-hour run to ~25 minutes

PROJECT EXPERIENCE

Engineering Physics Propulsion Laboratory: AIRHOUND [Middleware Lead]

September 2024 - May 2026

- Designed a modular four-node ROS2 perception-to-control pipeline (detection, tracking, PINN prediction, PX4 bridge) communicating with a PX4 FMUv6X flight controller over Micro XRCE-DDS
- Diagnosed DDS image-transport serialization as the pipeline bottleneck and replaced it with direct V4L2 capture, cutting mean detection latency from 234 ms to 58.8 ms and eliminating all dropouts at 1280x720
- Deployed RF-DETR with TensorRT FP16 on a Jetson Orin at 15 FPS end-to-end, sustaining the 10 Hz PX4 offboard control loop

Dilithium: Post-Quantum Cryptocurrency

January 2026 - Present

- Built a peer-to-peer proof-of-work blockchain in Go (node, wallet CLI, CPU/GPU miner) with CRYSTALS-Dilithium (FIPS 204) post-quantum transaction signing
- Implemented competitive mining across a live P2P network with automatic block and transaction sync
- Exposed a REST API with UPnP port forwarding and cross-platform builds for Linux, macOS, and Windows

SEMBAS-RL: Graduate Research, Autonomous Driving Policy Assurance

June 2026 - Present

- Trained and evaluated fifteen RL driving policies (PPO and SAC) across five curriculum strategies over 8M+ training steps
- Extended SEMBAS, a three-phase adaptive boundary scanner, to RL policy evaluation over a 3D scenario space to characterize where a learned policy remains reliable
- Built three analysis agents that convert boundary geometry into ranked retraining proposals automatically

PUBLICATIONS

DiMario, C., Bacha, R., & Butka, B. "Combating Senior Scams Using a Large Language Model-Created Rubric." ASSE '24, ACM. doi:10.1145/3702138.3702140

Malarchick, R., DiMario, C., et al. "Predictive Target Pursuit for Autonomous UAVs Using RF-DETR with Depth-Aware State Estimation..." Proc. SPIE 14030 (2026). doi:10.1117/12.3094732

EXTRACURRICULAR ACTIVITIES

- **ERAU Division II Cross-Country/Track:** Approximately 30 hours of training weekly
- **ERAU Cyber Rodeo 2026:** First Place Individual

<https://dimario.dev>